Dimensionality Reduction — Key Facts

- **Dimension** = number of features/variables in a dataset.
- A dataset with 20 features is a **20-dimensional dataset**.
- High-dimensional data can be difficult to **visualize, store, process, and model**.
- High dimensionality can increase **computational cost**.
- High-dimensional datasets may contain **redundant or correlated features**.
- Dimensionality reduction means representing data using **fewer dimensions**.

  Example: **100 features → 10 dimensions**.
- Dimensionality reduction can help with **visualization**.
- Dimensionality reduction can reduce **computational complexity**.
- Dimensionality reduction can reduce **storage requirements**.
- Dimensionality reduction may help reduce the effect of **noise and redundancy**.
- Dimensionality reduction does **not guarantee better model accuracy**.
- Reducing dimensions can result in **information loss**.
- The fewer dimensions we retain, the greater the potential information loss.
- There are two major approaches:
  - **Feature selection**
  - **Feature extraction**
- **Feature selection** keeps some of the original features.
- **Feature extraction** creates new features from the original features.

PCA — Key Facts

- **PCA = Principal Component Analysis**.
- PCA is an **unsupervised** technique.
- PCA does **not require class labels**.
- PCA is commonly used for **dimensionality reduction**.
- PCA is a **linear transformation** technique.
- PCA transforms the original features into **principal components**.
- Principal components are **new variables**.
- Principal components are **linear combinations of the original features**.
- PC1 is the direction with **maximum variance**.
- PC2 captures the **maximum remaining variance**.
- PC2 is **orthogonal to PC1**.

- PC3 is orthogonal to PC1 and PC2, and so on.
- Principal components are generally **uncorrelated** with each other.
- PCA orders components according to the amount of **variance they capture**.
- The first principal component captures the **largest amount of variance**.
- Later components generally capture progressively **less variance**.
- PCA can transform:

100 features → 100 principal components

- But if we keep only 10:

100 features → 10 principal components

then dimensionality has been reduced.

Variance & Explained Variance

- **Variance** measures the spread of data.
- PCA looks for directions with **high variance**.
- **Explained variance** tells us how much variance a principal component captures.
- If PC1 explains 70% and PC2 explains 20%, together they explain **90%**.
- **Cumulative explained variance** is the sum of explained variances up to a particular component.
- A common goal is to retain a chosen percentage such as **90%, 95%, or 99%** of the variance.
- More retained components → **more variance retained**.
- Fewer retained components → **greater dimensionality reduction**.
- More retained components generally → **lower reconstruction error**.
- Fewer retained components generally → **higher reconstruction error**.

Centering & Scaling

- PCA is commonly performed after **centering** the data.
- Centering means subtracting the **mean** from each feature.
- After centering, each feature has approximately **zero mean**.
- Scaling is important when features have **very different units or magnitudes**.
- Example:
    - Age → 20–60
    - Income → 20,000–200,000
- Without scaling, a large-scale feature may have a **larger influence on variance**.

- Standardization commonly produces features with:
    - Mean ≈ **0**
    - Standard deviation ≈ **1**

Covariance & PCA

- PCA is commonly derived using the **covariance matrix**.
- Covariance describes how two variables **vary together**.
- Positive covariance → variables tend to increase together.
- Negative covariance → one tends to increase while the other decreases.
- Highly correlated features often contain **redundant information**.
- PCA can represent correlated information using **fewer principal components**.

Eigenvalues & Eigenvectors

- PCA can be calculated using **eigenvalues and eigenvectors** of the covariance matrix.
- **Eigenvectors → principal directions**.
- **Eigenvalues → variance associated with those directions**.
- Larger eigenvalue → **more variance captured**.
- Smaller eigenvalue → **less variance captured**.
- Principal components are usually arranged in **decreasing eigenvalue order**.
- The eigenvector corresponding to the largest eigenvalue gives **PC1**.

PCA and Information

- PCA tries to preserve **variance**.
- PCA does **not necessarily preserve task-specific information**.
- High variance does not always mean **high predictive importance**.
- PCA does not directly maximize **classification accuracy**.
- PCA does not directly maximize **class separation**.
- Therefore, PCA does **not guarantee improved classification performance**.
- If important class information exists in a low-variance direction, removing that component may hurt classification.

PCA & Visualization

- PCA can reduce high-dimensional data to **2D or 3D**.

- PC1 and PC2 can be plotted to visualize data.
- PCA plots can help identify:
    - Clusters
    - Trends
    - Outliers
    - Patterns
    - Possible class separation
- A 2D PCA plot is only a **projection** of the original data.
- Important information may exist in components that are not plotted.

Feature Selection vs PCA

- Feature selection:

  Original features → select some original features

- PCA:

  Original features → create new principal components

- If you select:

  Age, Height, Weight

  these remain the original variables.

- PCA might instead produce:

  PC1, PC2, PC3

  where each PC is a combination of multiple original variables.

PCA Reconstruction

- PCA can be used to create a lower-dimensional representation.
- The lower-dimensional representation can be used to approximately reconstruct the original data.
- Keeping all components retains the complete dimensional representation.
- Keeping only some components produces an approximation.
- Reconstruction error generally decreases as more components are retained.
- Dimensionality reduction therefore involves a trade-off between compression and information retention.